Abhishek Mallick

CONTACT INFORMATION

Dartmouth College

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RESEARCH INTERESTS

Low dimensional topology. Heegaard Floer homology. Seiberg-Witten Theory. Instanton Floer Homology. Khovanov homology.

EMPLOYMENT

Dartmouth College

Assistant Professor (tenure-track), 2025-Current.

Rutgers University, New Brunswick

Hill Assistant Professor, 2023 Spring-2025 Summer.

Mathematical Sciences Research Institute (SLMath), Berkeley

Postdoctoral Fellow Fall, 2022

Max-Planck-Institut für Mathematik, Bonn

Postdoctoral Research Fellow 2021-22

EDUCATION

Michigan State University

Ph.D. Mathematics 2021

- Advisor: Matthew Hedden

Ramakrishna Mission Vivekananda Educational and Research Institute, India

M.Sc. in Mathematics, 2015

Ramakrishna Mission Vidyamandira, India

B.Sc. in Mathematics, 2013

EXTENDED VISITS

Junior Scientific Leader, for the Simons Semester, 2024

Institute of Mathematics at the Polish Academy of Sciences, Poland.

PUBLICATIONS

Corks, involutions, and Heegaard Floer Homology, *Journal of the European Mathematical Society*, 25(6) (2023), 2319-2389. (with Dai and Hedden).

Equivariant knots and knot Floer homology. *Journal of Topology*, 16(3), (2023), 1167-1236 (with Dai and Stoffregen).

Knot Floer homology and surgery on equivariant knots, *Journal of Topology* 17.4 (2024): e70001.

The (2,1)-cable of the figure-eight knot is not smoothly slice, *Inventiones Mathematicae* 238.2 (2024): 371-390 (with Dai, Kang, Park, and Stoffregen).

Rank-expanding satellites, Whitehead doubles, and Heegaard Floer homology, (with Dai, Hedden, and Stoffregen), *Journal of Topology* 17.4 (2024): e70008.

From diffeomorphisms to exotic phenomena in small 4-manifolds, arXiv: 2304.05997 (with Konno, and Taniguchi) to appear in *Selecta Mathematica*.

Exotic Dehn twists on 4-manifolds, arXiv: 2306.08607 (with Konno, and Taniguchi) to appear in *Geometry & Topology*.

Involutions and the Chern–Simons filtration in instanton Floer homology, arXiv: 2309.02309 (with Alfieri, Dai, and Taniguchi) to appear in the *Journal of Differential Geometry*.

Gompf's cork and Heegaard Floer homology, *International Mathematical Research Notices*, 2024, 18, 12663-12682 (with Dai and Zemke).

On localizing groups of exotic diffeomorphisms of 4-manifolds, arXiv: 2406.11773 (with Konno) to appear in the *Mathematische Annalen*.

A note on cables and the involutive concordance invariants, arXiv: 2409.02192 (with Hendricks) *Bulletin of the London Mathematical Society* 57.5 (2025): 1593-1604.

Exotically knotted closed surfaces from Donaldson's diagonalization for families, arXiv: 2409.07287 (with Konno, and Taniguchi).

The link surgery formula and equivariant surgeries, arXiv: 2507.12809 (with Hendricks, Stoffregen and Zemke).

Khovanov homology and equivariant surfaces, arXiv: 2507.13642 (with Borodzik, Dai and Stoffregen) to appear in the *Advances in Mathematics*.

On absolutely exotic diffeomorphisms of 4-manifolds, arXiv: 2510.05505 (with Konno, and Taniguchi).

Reidemeister and movie moves for involutive links, arXiv: 2604.26369 (with Borodzik, Dai and Stoffregen)).

Exotic disks and singular instanton Floer homology, arXiv: 2606.05819 (with Dai, and Taniguchi).

Fellowships and Awards

09/26-08/29	NSF Topology grant DMS-2604218 ($317,914).
2026-2031	Simons Foundation's travel award, TSM-00025903 ($35,000).
2025	The 2025 Frontiers of Science Award, International Congress for Basic Science.
2025	Burke Research Initiation Award, Dartmouth College.
2025	Titchmarsh Research Fellowship, University of Oxford (declined).
2025	Chapman Fellowship, Imperial College London (declined).
2025	AIM SQUARE, Caltech, with Irving Dai, Maggie Miller and Matt Stoffregen.

2020	Douglas A. Spragg Endowed Fellowship in Mathematics, MSU.
2020	College of Natural Science Dissertation completion Fellowship, MSU.
2018	Paul and Wilma Dressel Endowed Scholarship, MSU.
2016	College of Natural Science Dissertation Continuing Fellowship, MSU.
2016	Paul and Wilma Dressel Endowed Scholarship, MSU.
2010–2015	Innovation in Science Pursuit for Inspired Research Fellowship, Department of Science and Technology, Govt.of India.
2010–2015	Jagadis Bose National Science Talent Search Scholarship.

Invited Talks & Visits

University of Texas at Austin, Topology Seminar, 2026 (upcoming)

Recent advances in low-dimensional topology, 2027 at ICTS, India. (upcoming)

New structures in low-dimensional topology workshop, Mini-course, AIM, 2026

Colloquium, Tata Institute For Fundamental Research, Mumbai, India, 2026 (upcoming)

Graduate Student Geometry and Topology Conference, Rutgers University - New Brunswick, 2026

AMS Special Session on Topology, Gauge Theory, and Special Geometric Structures, Boston College, 2026

University of Massachusetts, Amherst, Topology Seminar, 2026

Michigan State University, Topology Seminar, 2025

Columbia University, Topology Seminar, 2025

Boston College, Topology Seminar, 2025

Canadian Mathematical Society, 2025 Summer Meeting, Low-dimensional Topology session (unable to travel)

Simons Center's Workshop: Gauge Theory and Floer Homology in Low Dimensional Topology, 2025

University of Michigan, Topology Seminar, 2025

Gauge Theory Virtual Seminar, 2025

Rutgers Undergraduate Mathematical Association Seminar, 2024

University of Texas at Austin, 2024

Link invariants and surfaces in 4-manifolds, AMS special session, University of Texas, San Antonio, 2024

Braids Reunion Workshop, ICERM, Brown University, 2024

Institute of Mathematics of the Polish Academy of Sciences, Simons Semester-Knots, Homologies and Physics, 2024

University of British Columbia, Topology seminar, 2024 (unable to travel)

MIT, Geometry and Topology seminar, 2024

Gauge theory learning seminar, Rutgers University, 2024

UCLA, 2024 Geometry and Topology Workshop UCLA, 2024

Stony Brook University, Symplectic Geometry, Gauge Theory and Low-Dimensional Topology seminar, 2023

Columbia University, Geometric Topology seminar, 2023

University of Virginia, New Developments in 3- and 4-Manifold Topology, 2023.

University of Georgia, Geometry and Topology seminar, 2023.

Log Cabin Conference on Concordance and Knotted Surfaces, Arizona, 2023.

Rutgers University - New Brunswick, Geometry and Topology seminar, 2023.

Oberwolfach Workshop: Morphism in Low-dimensional topology, 2023 (unable to attend)

MIT, Geometry and Topology seminar, 2022.

Stanford University, Topology seminar, 2022.

MSRI, Berkeley, Floer homotopy theory program seminar, 2022.

Princeton University, Topology seminar, 2022.

IBS Center for Geometry and Physics, CGP seminar, 2022.

Max Planck Institute for Mathematics, Surfaces in 4-manifolds, 2022.

American Institute of Mathematics, Program on 4-manifolds, virtual, 2021.

American Mathematical Society, special session on the Topology and Geometry of 3- and 4-manifolds, AMS Southeastern Sectional Meeting, (virtual) Georgia Tech, 2021.

Joint Mathematics Meetings: AMS Special Session on Low Dimensional Topology, (Associated with AMS Invited Maryam Mirzakhani Lecture), virtual conference, 2021.

Nearly Carbon Neutral Geometry Topology Conference, mini-session on 4-manifolds, virtual conference, 2020.

American Mathematical Society, Sectional Meeting; Special Session on Low-dimensional Topology, Purdue University (canceled), 2020.

University of Virginia, Geometry Seminar, 2020.

Graduate Student Topology and Geometry Conference, Indiana University Bloomington (postponed), 2020.

Undergraduate student supervision

Jay Patwardhan - Rutgers University (REU), Zheheng Xiao - Columbia University (REU).

Ribhu Hooja, Sean Kim, Filip Rupchin, Summer Hybrid Undergraduate Research Program, Dartmouth College, 2026

Henry Dorr - Dartmouth College, Summer Reading Course, 2026.

Graduate student supervision

Allison Tsypin, Dartmouth College.

Professional service

Member of *Graduate Admission Committee*, Dartmouth College, Department of Mathematics, 2025

Referee/Quick opinion for *Advances in Mathematics*, *Algebraic & Geometric Topology*, *Compositio Mathematica*, *Geometry & Topology*, *International Mathematics Research Notices*, *Journal of the European Mathematical Society*, *Journal of Topology*, *K3 problem list*, *Mathematische Annalen*, *Quantum Topology*, *Selecta Mathematica*, *Transactions of the American Mathematical Society.*

Mentor for an REU project (Rutgers University, Summer 2023) on the topic *Generalized Mazur pattern and Bordered Heegaard Floer homology.*

Faculty Mentor for Summer Hybrid Undergraduate Research Program, Dartmouth College.

Teaching

Spring	2026	Primary Instructor, *Graduate Algebraic Topology*
Winter	2026	Primary Instructor, *Current Problems in Topology* (graduate topics course)
Spring	2025	Primary Instructor, Introduction to Mathematical Reasoning
Fall	2024	Primary Instructor, two sections of Calculus I (Differentiation and Integration)
Spring	2024	Primary Instructor, *Floer Homology and Low-Dimensional Topology*, Graduate Topics Course
Fall	2023	Primary Instructor, two sections of Calculus I (Differentiation and Integration)

Spring	2023	Primary Instructor, two sections of Calculus I (Differentiation and Integration)
Spring	2020	Teaching Assistant, Calculus III (Multivariable Calculus)
Fall	2019	Teaching Assistant, Calculus III (Multivariable Calculus)
Summer	2019	Primary Instructor, College Algebra
Spring	2019	Teaching Assistant, Graduate course on Algebraic Topology
Fall	2018	Teaching Assistant, Calculus III (Multivariable Calculus)
Spring	2018	Teaching Assistant, Calculus III (Multivariable Calculus)
Fall	2017	Teaching Assistant, Transition to Proofs
Summer	2017	Primary Instructor, Calculus IV (Differential Equations)
Spring	2017	Primary Instructor, Calculus IV (Differential Equations)
Fall	2016	Primary Instructor, Calculus II (Integration, Series, Sequences)
Summer	2016	Primary Instructor, Calculus II (Integration, Series, Sequences)

Teaching Organization

Co-organizing Dartmouth Topology seminar 2025-

Co-organizing a Special Session in Joint Mathematics Meetings (JMM), 2025- 'Concordance and Cobordism in Low-dimensions' with Maggie Miller and Ryan Stees

Co-organizing Rutgers Geometry and Topology weekly seminar, 2024

Co-organized Rutgers Gauge Theory, Low-Dimensional Topology, and Geometric Analysis Conference, 2024

Co-organized Rutgers DIMACS REU, Summer 2023

Co-organized Postdoctoral Research Seminar on Floer homotopy theory, MSRI-SLMath, 2022

Co organized a mini-session in Nearly Carbon Neutral Geometry Topology Conference, virtual conference, 2021.

Co-organized weekly student Geometry and Topology seminar, MSU, 2017-2018.

Co-organized Graduate Student Topology and Geometry Conference, MSU, 2017.